State Estimation and Filtering for Robotic Perception

AA273

Kalman Filter independence assumptions, Cond. Indep. Measurements, Markovianity



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 3 due Fri Jan 30
- HW 4 out Fri Jan 30, due Fri Feb 13
- Group signup sheet posted (groups from 1 – 4 people)
- Midterm: Feb 5-7
  - 5 hour timed take home exam, you choose any 5 hours in 48 hour window
  - Open book, notes, lecture videos
  - No internet, gen AI, or collaboration, coding assistants are allowed.
- Project Proposal: due Tue Feb 17
  - 2-3 pages, Latex
  - Related work >= 10 refs, ~1 page
  - Proposed work ~ 1 page
  - Bibtex refs

Lecture 8:

- Kalman Filter independence assumptions
- Cond. Indep. Measurements, Markovianity

Recall: L-G SS Model and KF Eqns

Dynamics eqn.:

$$X_{t+1} = A_t X_t + B_t u_t + W_t$$

Measurement eqn.:

$$Y_t = C_t X_t + D_t u_t + V_t$$

Initial condition:

$$X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$$

Whiteness and Indep.:

$$W_t \sim GWN(0, Q_t)$$

$$V_t \sim GWN(0, R_t)$$

X_0, W_t, V_τ independent for all t, τ

Given: $\mu_{0|0}, \Sigma_{0|0},\ A_t, B_t, C_t,\ Q_t, R_t$

(Assume $D_t = 0$)

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop

Recall: Assumptions in KF Derivation

Assumptions we made in deriving KF:

A1: V_t indep. of $Y_{1:t-1}$

A2: X_t indep. of V_t given $Y_{1:t-1}$

A3: W_t indep. of $Y_{1:t}$

A4: W_t indep. of X_t given $Y_{1:t}$

Whiteness and Indep. Properties of L-G SS model:

M1: $W_t \sim GWN(0, Q_t)$

M2: $V_t \sim GWN(0, R_t)$

M3: X_0, W_t, V_τ independent for all t, τ

Claim 1: M.1 – M.3 imply A1 – A4.

Claim 2: L-G SS Model with M.1-M.3 is Markov and has cond. Independent measurements.

Note: Claim 1 completes the proof of correctness of the KF for the L-G SS model. Claim 2 is essential to the existence of a recursive Bayesian filter for the L-G SS Model.

Intermediate Simplification

Recall HW 3 Prob 2: If $(X_t, Y_{1:t})$ indep of W_t, then $Cov(X_t, W_t \mid Y_{1:t}) = 0$.

For $(X_t, W_t, Y_{1:t})$ jointly Gaussian, this implies A4.

Same reasoning: If $(X_t, Y_{1:t-1})$ indep of V_t, then $Cov(X_t, V_t \mid Y_{1:t-1}) = 0$.

For $(X_t, V_t, Y_{1:t-1})$ jointly Gaussian, this implies A3.

A1: $Y_{1:t-1}$ indep. of V_t

A2: X_t indep. of V_t given $Y_{1:t-1}$

A3: $Y_{1:t}$ indep. of W_t

A4: X_t indep. of W_t given $Y_{1:t}$





A1: $Y_{1:t-1}$ indep. of V_t

A2': X_t indep. of V_t

A3: $Y_{1:t}$ indep. of W_t

A4': X_t indep. of W_t

Recall: Assumptions in KF Derivation

Assumptions we made in deriving KF:

A1: $Y_{1:t-1}$ indep. of V_t

A2': X_t indep. of V_t

A3: $Y_{1:t}$ indep. of W_t

A4': X_t indep. of W_t

Whiteness and Indep. Properties of L-G SS model:

M1: $W_t \sim GWN(0, Q_t)$

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Prove A4': X_t indep. of W_t

$$
\begin{aligned}
X_t &= A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1} \\
&= A_{t-1}A_{t-2}X_{t-2} + A_{t-1}B_{t-2}u_{t-2} + B_{t-1}u_{t-1} + A_{t-1}W_{t-2} + W_{t-1} \\
&\vdots \quad \text{Recurse to t = 0.} \\
&= \left(\prod_{\tau=0}^{t-1} A_\tau\right) X_0 + \sum_{\tau=0}^{t-1}\left(\prod_{s=\tau+1}^{t-1} A_s\right) B_\tau u_\tau + \sum_{\tau=0}^{t-1}\left(\prod_{s=\tau+1}^{t-1} A_s\right) W_\tau \\
&= \Phi_{t-1}X_0 + \underbrace{\begin{bmatrix}\Psi_{t-1} & \cdots & \Psi_0\end{bmatrix}}\begin{bmatrix} B_{t-1}u_{t-1} \\ \vdots \\ B_0 u_0 \end{bmatrix} + \underbrace{\begin{bmatrix}\Psi_{t-1} & \cdots & \Psi_0\end{bmatrix}}\begin{bmatrix} W_{t-1} \\ \vdots \\ W_0 \end{bmatrix}
\end{aligned}
$$

Prove A4': X_t indep. of W_t

$$X_t = \Phi_{t-1} X_0 + \begin{bmatrix} \Psi_{t-1} & \cdots & \Psi_0 \end{bmatrix} \begin{bmatrix} B_{t-1} u_{t-1} \\ \vdots \\ B_0 u_0 \end{bmatrix} + \begin{bmatrix} \Psi_{t-1} & \cdots & \Psi_0 \end{bmatrix} \begin{bmatrix} W_{t-1} \\ \vdots \\ W_0 \end{bmatrix}$$

X_t only depends on RVs W_τ for all $\tau < t$ and X_0

A4' ⟸

- So W_τ indep. of W_t for all $\tau < t$ ⟸ M1: $W_t \sim GWN(0, Q_t)$
- And X_0 indep. of W_t for all t ⟸ M3: X_0, W_t, V_τ indep. for all t, τ

Prove A3: $Y_{1:t}$ indep. of W_t

$$Y_\tau = C_\tau X_\tau + D_\tau u_\tau + V_\tau$$

Y_τ only depends on RVs X_τ and V_τ for all $\tau = 1, \ldots, t$

A3 ⟵

So X_τ indep. of W_t for all $\tau \leq t$ ⟵ A4': X_t indep. of W_t

And V_τ indep. of W_t for all $\tau \leq t$ ⟵ M3: X_0, W_t, V_τ indep. for all t, τ

Prove A2': X_t indep. of V_t

$$X_t = \Phi_{t-1} X_0 + \begin{bmatrix} \Psi_{t-1} & \cdots & \Psi_0 \end{bmatrix} \begin{bmatrix} B_{t-1} u_{t-1} \\ \vdots \\ B_0 u_0 \end{bmatrix} + \begin{bmatrix} \Psi_{t-1} & \cdots & \Psi_0 \end{bmatrix} \begin{bmatrix} W_{t-1} \\ \vdots \\ W_0 \end{bmatrix}$$

On the board and in the notes.

Prove A1: $Y_{1:t-1}$ indep. of V_t

$$X_t = \Phi_{t-1} X_0 + \begin{bmatrix} \Psi_{t-1} & \cdots & \Psi_0 \end{bmatrix} \begin{bmatrix} B_{t-1} u_{t-1} \\ \vdots \\ B_0 u_0 \end{bmatrix} + \begin{bmatrix} \Psi_{t-1} & \cdots & \Psi_0 \end{bmatrix} \begin{bmatrix} W_{t-1} \\ \vdots \\ W_0 \end{bmatrix}$$

$$Y_\tau = C_\tau X_\tau + D_\tau u_\tau + V_\tau$$

On the board and in the notes.

Linear Structure of Jointly Gaussian RVs

On the board and in the notes.